

Skills Summary

Naming the Angle Pairs a Transversal Makes

Use this reference while completing your worksheet or when studying.

Skill 1 — Reading the picture and naming the pair

Two words locate every pair. **Interior** means between the two lines; **exterior** means outside them. **Alternate** means opposite sides of the transversal; **same-side** means the same side.

- Corresponding — one interior, one exterior, same side: *congruent*
- Alternate interior, alternate exterior: *congruent*
- Same-side interior: *supplementary*

These three lines need the two lines to be **parallel**. Vertical angles and linear pairs do not — they hold at any crossing.

Example: $j \parallel k$ and $m\angle 4 = 112^\circ$, where $\angle 4$ and $\angle 6$ are same-side interior angles. Find $m\angle 6$.

Step 1. Both angles sit between the lines and on the same side of the transversal, so the pair is same-side interior — and that name says supplementary, so I subtract from 180 instead of copying the measure.

Step 2. $m\angle 6 = 180 - 112$.

$\Rightarrow$ **68** degrees

Try it: $j \parallel k$ and one same-side interior angle measures 133° . Find the other.

check: 47

Skill 2 — Turning the name into an equation

Congruent pair (corresponding, alternate interior, alternate exterior): set the two expressions **equal**.

Supplementary pair (same-side interior, and any linear pair): **add** the two expressions and set the sum to 180.

Write the name down before writing the equation. Choosing “equal” for a same-side pair is the most common error on this topic, and the arithmetic afterwards hides it.

Example: $j \parallel k$, and corresponding angles measure $(2x + 35)^\circ$ and $(4x - 15)^\circ$. Find x .

Step 1. The pair is corresponding and the lines are parallel, so the two angles are congruent — which makes the equation an equality of the two expressions, not a sum.

Step 2. $2x + 35 = 4x - 15$, so $50 = 2x$.

$\Rightarrow$ **$x = 25$**

Try it: Alternate interior angles measure $(3x + 12)^\circ$ and $(5x - 28)^\circ$. Find x .

check: $07 = x$

Skill 3 — Chaining two relationships

When the two angles are not a named pair, travel through a third angle: use a **corresponding** or **vertical** angle to move to the other intersection, then a **linear pair** to move around that vertex.

Supplementary chain: if a same-side interior pair is $(a)^\circ$ and $(b)^\circ$, then $a + b = 180$; solve for the unknown, then substitute back to get each measure.

Example: $j \parallel k$, and a same-side interior pair measures $(5x + 10)^\circ$ and $(3x + 26)^\circ$. Find x .

Step 1. The pair is same-side interior, so the two measures add to 180; that single equation carries both expressions and the one unknown.

Step 2. $(5x + 10) + (3x + 26) = 180$, so $8x + 36 = 180$ and $8x = 144$.

$$\Rightarrow x = 18$$

Try it: A same-side interior pair measures $(7x + 4)^\circ$ and $(2x + 5)^\circ$. Find x .

61 = x :check

(!) **Watch out:** no parallel marks, no parallel rules. Without them only vertical angles and linear pairs are guaranteed.